[ ]:
```
Loops : a loop is a programming concept used to excute a block of code repeatedly until
        certain condition is met or sequence is completed
types of loops
.for loop
   1.for loop with sequence
   2.for loop with range
.while loop
.nested loop
   1.for loop inside for loop
   2.for loop inside while loop
   3.while loop inside while loop
   4.while loop inside for loop
```

[3]:
```
#for loop
name = "jungkook jeonghan"
for char in name:
    print(char,end=" ")
```

```
j u n g k o o k   j e o n g h a n
```

[4]:
```
#position of each character
name="jeonghan"
pos=0
for char in name:
    print(f"position : {pos},character : {char}")
    pos =pos+1
```

```
position : 0,character : j
position : 1,character : e
position : 2,character : o
position : 3,character : n
position : 4,character : g
position : 5,character : h
position : 6,character : a
position : 7,character : n
```

[6]:
```
for i in range(1,11,1):
    print(f"good morning : {i}")
```

```
good morning : 1
good morning : 2
good morning : 3
good morning : 4
good morning : 5
good morning : 6
good morning : 7
good morning : 8
good morning : 9
good morning : 10
```

[1]:
```
lis = ['a','b','c','d','e']
for i in lis:
    print("good morning",i)
```

```
good morning a
good morning b
good morning c
good morning d
good morning e
```

[4]:
```
name = "jungkook"
for char in name:
    print(char,end=" ")
```

```
j u n g k o o k
```

[5]:
```
name ="jeonghan"
for char in name:
    print(char,end="\t")
```

```
j	e	o	n	g	h	a	n
```

[7]:
```
name = "taehung"
pos = 0
for char in name:
    print("position : ",pos,"character is : ",char)
    pos=pos+1
```

```
position :  0 character is :  t
position :  1 character is :  a
position :  2 character is :  e
position :  3 character is :  h
position :  4 character is :  u
position :  5 character is :  n
position :  6 character is :  g
```

[9]:
```
name = "taehung"
pos = 0
for char in enumerate(name):
    print(f"position {pos} character: {char} ")
    pos=pos+1
```

```
position 0 character: (0, 't')
position 1 character: (1, 'a')
position 2 character: (2, 'e')
position 3 character: (3, 'h')
position 4 character: (4, 'u')
position 5 character: (5, 'n')
position 6 character: (6, 'g')
```

[15]:
```
l1 = ["jk","han","v","jin","jimin","suga","hopw","rm","scoups","dk","hoshi"]
for i in l1:
    print(f"happy diwali {i}  have a bright future")
```

```
happy diwali jk      have a bright future
happy diwali han      have a bright future
happy diwali v      have a bright future
happy diwali jin      have a bright future
happy diwali jimin      have a bright future
happy diwali suga      have a bright future
happy diwali hopw      have a bright future
happy diwali rm      have a bright future
happy diwali scoups      have a bright future
happy diwali dk      have a bright future
happy diwali hoshi      have a bright future
```

[ ]:
```
for loop with range intialization/start value condition/stop value increment/decrement/step size

range(start value,stop value,step size)
default start value= 0
stop value = upto given condition
step size =1

for iterative in rane(stv,spv,ss):
    statements
```

[17]:
```
for i in range(1,5,1):
    print("Hi! good morning")
```

```
Hi! good morning
Hi! good morning
Hi! good morning
Hi! good morning
```

[9]:
```
name = "taehung"
pos = 0
for char in enumerate(name):
    print(f"position {pos} character: {char} ")
    pos=pos+1
```

```
position 0 character: (0, 't')
position 1 character: (1, 'a')
position 2 character: (2, 'e')
position 3 character: (3, 'h')
position 4 character: (4, 'u')
position 5 character: (5, 'n')
position 6 character: (6, 'g')
```

[15]:
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```

```
happy diwali jk      have a bright future
happy diwali han      have a bright future
happy diwali v      have a bright future
happy diwali jin      have a bright future
happy diwali jimin      have a bright future
happy diwali suga      have a bright future
happy diwali hopw      have a bright future
happy diwali rm      have a bright future
happy diwali scoups      have a bright future
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[ ]:
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```

```
Hi! good morning
Hi! good morning
Hi! good morning
Hi! good morning
```

[18]:
```
#sum of 1st 10 numbers
sum =0
for i in range(1,11):
    sum+=i
print(sum)
```

```
55
```

[11]:
```
#factorial of a number
n = int(input("Enter a number"))
fact = 1

for i in range(n,0,-1):
    fact = fact*i
print(f"factorial of {n} is {fact}")
```

```
Enter a number 5
factorial of 5 is 120
```

[19]:
```
#number is prime or not
num = int(input("Enter  a number "))
m = 0
for i in range(num,1,-1):
    if(num%i==0):
        m+=1
if(m==1):
    print(f"{num} is prime number")
else:
    print(f"{num} is not prime number")
```

```
Enter  a number  23
23 is prime number
```

[ ]:
```
loop statements
break
continue
pass
```

[20]:
```
#break
for i in range(1,100,1):
    if(i==5):
        break
    else:
        print(i)
```

```
1
2
3
4
```

[21]:
```
#continue
for i in range(1,10,1):
    if(i==5):
        continue
    else:
        print(i)
```

```
1
2
3
4
6
7
8
9
```

[25]:
```
#number is prime or not
num = int(input("Enter  a number "))
for i in range(2,num,1):
    if(num%i==0):
        print("its not a prime number")
        break
else:
    print("prime number")
```

```
Enter  a number  15
its not a prime number
```

[30]:
```
#print a table
num = int(input("Enter a number"))
for i in range(1, 11,1):
    print(num,"x" ,i ,"=",num*i)
```

```
Enter a number 8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
```